CSC3105

Data Analytics and AI Revision

Naïve Bayes

Consider the 10 training data set with feature A, B and C given below

Instance	A	B	C	Class Label, L
1	0	0	0	1
2	0	0	1	0
3	0	1	1	0
4	0	1	1	0
5	0	0	1	1
6	1	0	1	1
7	1	0	1	0
8	1	0	1	0
9	1	1	1	1
10	1	0	1	1

a) Estimate the conditional probabilities for $P(A=1|L=1), P(B=1|L=1), P(C=1|L=1), P(A=1|L=0), P(B=1|L=0), P(C=1|L=0), P(L=1), P(L=0)$

<u>Solution :</u>

$$P(A=1|L=1)=\frac{3}{5}=0.6, P(B=1|L=1)=\frac{1}{5}=0.2, P(C=1|L=1)=\frac{4}{5}=0.8$$

$$P(A=1|L=0)=\frac{2}{5}=0.4, P(B=1|L=0)=\frac{2}{5}=0.4, P(C=1|L=0)=\frac{5}{5}=1$$

$$P(L=1)=\frac{5}{10}=0.5, P(L=0)=\frac{5}{10}=0.5$$

b) Use the estimate of conditional probabilities given in the previous question to predict the class label for a test sample $(A=0, B=1, C=0)$ using the naive Bayes approach.

<u>Solution :</u>

Let $P(A=0, B=1, C=0)=D$

Using Naïve Bayes,

$$P(L=1|A=0,B=1,C=0)=\frac{P(L=1,A=0,B=1,C=0)}{P(A=0,B=1,C=0)}$$
$$=\frac{P(A=0,B=1,C=0|L=1)P(L=1)}{P(A=0,B=1,C=0)}$$

Assume conditional independency that is every feature $A, B\ and\ C$ is independent of each other conditioned on L

$$P(L = 1|A = 0, B = 1, C = 0) = \\ = \frac{P(A = 0|L = 1)P(B = 1|L = 1)P(C = 0|L = 1)P(L = 1)}{P(A = 0, B = 1, C = 0)} \\ = \frac{2/5 \times 0.2 \times 1/5 \times 5/10}{P(A = 0, B = 1, C = 0)} = \frac{0.008}{P(A = 0, B = 1, C = 0)}$$

Similarly,

$$P(L = 0|A = 0, B = 1, C = 0) = \\ = \frac{P(A = 0|L = 0)P(B = 1|L = 0)P(C = 0|L = 0)P(L = 0)}{P(A = 0, B = 1, C = 0)} \\ = \frac{3/5 \times 2/5 \times 0/5 \times 5/10}{P(A = 0, B = 1, C = 0)} = \frac{0}{P(A = 0, B = 1, C = 0)} = 0$$

Therefore since, $P(L = 1|A = 0, B = 1, C = 0) > P(L = 0|A = 0, B = 1, C = 0)$

The class label will be $L = 1$

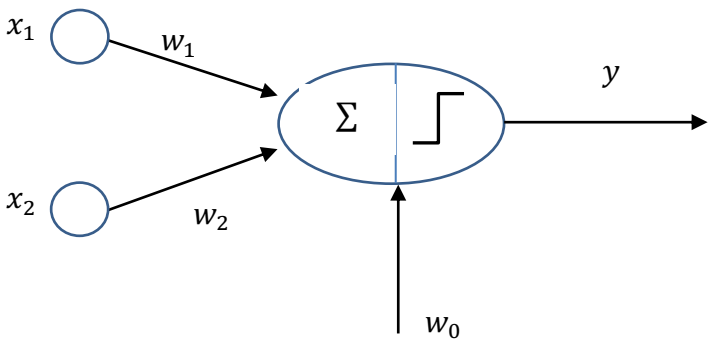
Multi-Layer Perceptron (MLP)

For a training dataset where x_1 and x_2 are the features and y is the class label as followed

x_1	x_2	y
0	0	1
0	1	0
1	0	0
1	1	0

a) Construct a perceptron to realise above classification and plot the scatterplot for above data point.

<u>Solution :</u>



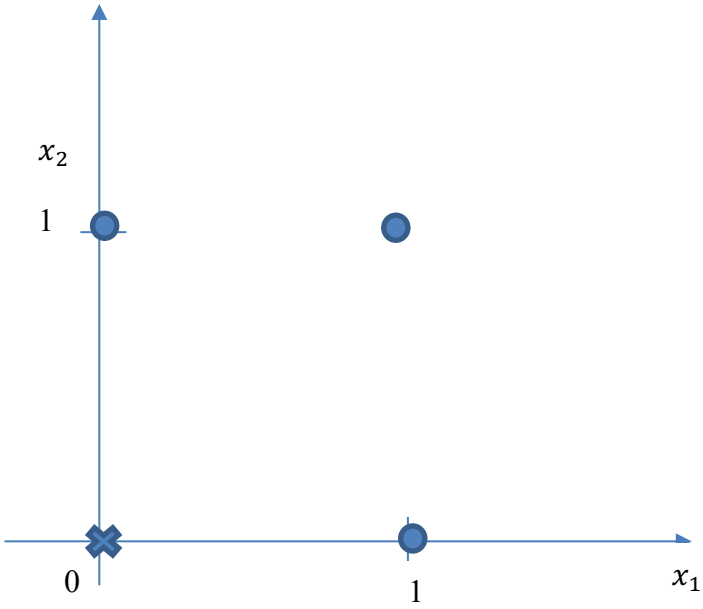
Find the equation of the predicted y , $\hat{y}$

$\hat{y} = w_1x_1 + w_2x_2 + w_0 \ = \mathbf{w}^{\mathbf{T}}\mathbf{x}$ where $\mathbf{w} = [w_1 \quad w_2 \quad w_0\,]^T, \mathbf{x} = [x_1 \quad x_2 \quad 1]^T$

Therefore

$$\hat{y} = sign(\mathbf{w}^{\mathbf{T}}\mathbf{x})$$

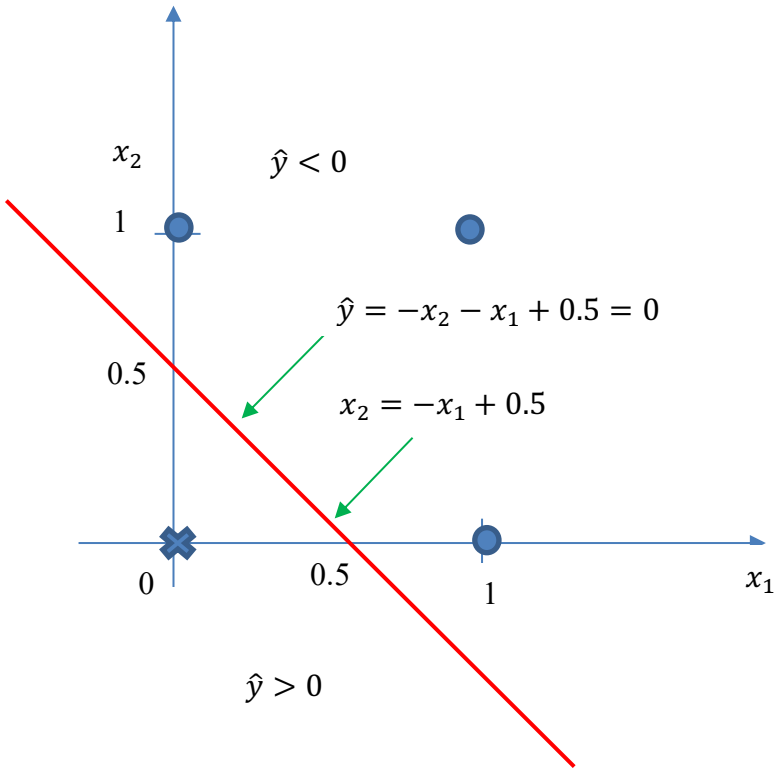
where sign is the activation function.



Circle represent class label 0 while x represent class label 1

b) Obtain the respective weight of the perceptron to realise the classification. Hence , plot the decision line for the classification.

Solution :



From the above

the line $x_2 = -x_1 + 0.5$ is the decision line where it can be transformed into

$$\hat{y} = w_1 x_1 + w_2 x_2 + w_0$$

$$= -x_1 - x_2 + 0.5$$

where $w_1 = -1, w_2 = -1$ and $w_0 = 0.5$. And it matches the truth table, so that any value of $\hat{y} > 0$ will be class 1 and $\hat{y} < 0$ will be class 0 with any combination x_1 and x_2,

Gradient Descent for Multi Features

For the training dataset where x_1 and x_2 are the features and y is the class label as followed

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

a) Using the back propagation gradient descent approach, the initial predicted decision line of perceptron for the above dataset is given $\hat{y} = w_1x_1 + w_2x_2 + w_o = 0.x_1 + 0x_2 - 2$ with an estimated error of -1. Obtain the final weight of the two features and the bias that will result in minimum squared error prediction using back propagation gradient descent. The gradient descent approach is given as follows:

$$E(\mathbf{w}) = \sum_{i=1}^{4}(\hat{y}_i - y_i)^2$$
$$w_j^{k+1} = w_j^k - \lambda \frac{\partial E}{\partial w_j}$$

where E is the total error in prediction for all dataset, and y_i and $\hat{y}_i$ are the output for i^{th} dataset. w_j^t and w_j^{t+1} are the weights at time k and $k+1$ respectively and $j = 0,1,2$. Let the learning rate $\lambda = 0.5$.

<u>Solution :</u>

$$E(\mathbf{w}) = \sum_{i=1}^{4}(\hat{y}_i - y_i)^2$$

$$w_1^{k+1} = w_1^k - \lambda \frac{\partial E}{\partial w_1}$$

$$w_2^{k+1} = w_2^k - \lambda \frac{\partial E}{\partial w_2}$$

$$w_0^{k+1} = w_0^k - \lambda \frac{\partial E}{\partial w_0}$$

Let z=$(\hat{y}_i - y_i)$ and $\hat{y}_i = w_1x_{1i} + w_2x_{2i} + w_o$.
Using chain rule in differentiation where

$$\frac{\partial E}{\partial w_1} = \frac{\partial E}{\partial z}.\frac{\partial z}{\partial \hat{y}_i}.\frac{\partial \hat{y}_i}{\partial w_1} \;,\; \frac{\partial E}{\partial w_2} = \frac{\partial E}{\partial z}.\frac{\partial z}{\partial \hat{y}_i}.\frac{\partial \hat{y}_i}{\partial w_2} \text{ and } \frac{\partial E}{\partial w_0} = \frac{\partial E}{\partial z}.\frac{\partial z}{\partial \hat{y}_i}.\frac{\partial \hat{y}_i}{\partial w_0}$$

$$\frac{\partial E}{\partial w_1} = \frac{\partial}{\partial w_1}\sum_{i=1}^{4}\left(\hat{y}_i - y_i\right)^2 = \frac{\partial}{\partial w_1}\sum_{i=1}^{4} z^2$$

$$\frac{\partial E}{\partial z} = 2\sum_{i=1}^{4} z \;,\; \frac{\partial z}{\partial \hat{y}_i} = 1$$

$$\frac{\partial \hat{y}_i}{\partial w_1} = x_{1i} \;,\; \frac{\partial \hat{y}_i}{\partial w_1} = x_{2i} \;,\; \frac{\partial \hat{y}_i}{\partial w_0} = 1$$

The first order derivatives of the error with respect to the weight

Therefore, $\frac{\partial E}{\partial w_1} = \frac{\partial E}{\partial z} . \frac{\partial z}{\partial \hat{y}_i} . \frac{\partial \hat{y}_i}{\partial w_1} = (2 \sum_{i=1}^{4} z).1.x_{1i}$=$2\sum_{i=1}^{4}(\hat{y}_i - y_i)\, x_{1i}$

Therefore, $\frac{\partial E}{\partial w_2} = \frac{\partial E}{\partial z} . \frac{\partial z}{\partial \hat{y}_i} . \frac{\partial \hat{y}_i}{\partial w_2} = (2 \sum_{i=1}^{4} z).1.x_{2i}$=$2\sum_{i=1}^{4}(\hat{y}_i - y_i)\, x_{2i}$

Therefore, $\frac{\partial E}{\partial w_0} = \frac{\partial E}{\partial z} . \frac{\partial z}{\partial \hat{y}_i} . \frac{\partial \hat{y}_i}{\partial w_0} = (2 \sum_{i=1}^{4} z).1.1$=$2\sum_{i=1}^{4}(\hat{y}_i - y_i)$

Initial predicted decision line is

$\hat{y}_i = w_1 x_{1i} + w_2 x_{2i} + w_o = 0.x_{1i} + 0x_{2i} - 2$ with an estimated error of -1.

Assume at iteration 0,

$w_1 = 0 = w_1^0$
$w_2 = 0 = w_2^0$
$w_0 = -2 = w_0^0$
$E = -1$

Update of weight at iteration 0,

For binary available

If $\hat{y}_i = 0.x_{1i} + 0x_{2i} - 2 > 0 \rightarrow \hat{y}_i = 1$
If $\hat{y}_i = 0.x_{1i} + 0x_{2i} - 2 \leq 0 \rightarrow \hat{y}_i = 0$

For $i = 1$, $\hat{y}_1$=$0.0. + 0.0 - 2 = -2 \rightarrow \hat{y}_1 = 0$
For $i = 2$, $\hat{y}_2$=$0.0. + 0.1 - 2 = -2 \rightarrow \hat{y}_2 = 0$
For $i = 3$, $\hat{y}_3$=$0.1. + 0.0 - 2 = -2 \rightarrow \hat{y}_3 = 0$
For $i = 4$, $\hat{y}_3$=$0.1. + 0.1 - 2 = -2 \rightarrow \hat{y}_4 = 0$

$$\sum_{i=1}^{4}(\hat{y}_i - y_i)\, x_{1i} = [\hat{y}_1 - y_1].x_{11} + [\hat{y}_2 - y_2].x_{12} + [\hat{y}_3 - y_3].x_{13} + [\hat{y}_4 - y_4].x_{14}$$
$$= [0 - 0].0 + [0 - 0].0 + [0 - 0].1 + [0 - 1].1 = -1$$

$$\sum_{i=1}^{4}(\hat{y}_i - y_i)\, x_{2i} = [\hat{y}_1 - y_1].x_{21} + [\hat{y}_2 - y_2].x_{22} + [\hat{y}_3 - y_3].x_{23} + [\hat{y}_4 - y_4].x_{24}$$
$$= [0 - 0].0 + [0 - 0].1 + [0 - 0].0 + [0 - 1].1 = -1$$

$$\sum_{i=1}^{4}(\hat{y}_i - y_i) = [\hat{y}_1 - y_1] + [\hat{y}_2 - y_2] + [\hat{y}_3 - y_3] + [\hat{y}_4 - y_4]$$
$$= [0 - 0] + [0 - 0] + [0 - 0] + [0 - 1] = -1$$

$\frac{\partial E}{\partial w_1}$ =$2\sum_{i=1}^{4}(\hat{y}_i - y_i)\, x_{1i}$=2.-1=-2

$\frac{\partial E}{\partial w_2}$ =$2\sum_{i=1}^{4}(\hat{y}_i - y_i)\, x_{2i}$=2.-1=-2

$\frac{\partial E}{\partial w_0}$ =2$\sum_{i=1}^{3}(\hat{y}_i - y_i)$=2.-1=-2

$w_1^1 = w_1^0 - \lambda \frac{\partial E}{\partial w_1}$ =0-0.5*-2=1

$w_2^1 = w_2^0 - \lambda \frac{\partial E}{\partial w_1}$ =0-0.5*-2=1

$w_0^1 = w_0^0 - \lambda \frac{\partial E}{\partial w_1}$ =-2-0.5*-2=-1

Therefore, $\hat{y}_i = x_{1i} + x_{2i} - 1$. For iteration 1,

For $i = 1$, $\hat{y}_1$=$1.0. + 1.0 - 1 = 0 \rightarrow \hat{y}_1 = 0$
For $i = 2$, $\hat{y}_2$=$1.0. + 1.1 - 1 = 0 \rightarrow \hat{y}_2 = 0$
For $i = 3$, $\hat{y}_3$=$1.1. + 1.0 - 1 = 0 \rightarrow \hat{y}_3 = 0$
For $i = 4$, $\hat{y}_3$=$1.1. + 1.1 - 1 = 1 \rightarrow \hat{y}_4 = 1$
Hence, Total Error E= 0+0+0+0=0

$$\sum_{i=1}^{4}(\hat{y}_i - y_i)\, x_{1i} = [\hat{y}_1 - y_1].x_{11} + [\hat{y}_2 - y_2].x_{12} + [\hat{y}_3 - y_3].x_{13} + [\hat{y}_4 - y_4].x_{14}$$
$$= [0 - 0].0 + [0 - 0].0 + [0 - 0].1 + [1 - 1].1 = 0$$

$$\sum_{i=1}^{4}(\hat{y}_i - y_i)\, x_{2i} = [\hat{y}_1 - y_1].x_{21} + [\hat{y}_2 - y_2].x_{22} + [\hat{y}_3 - y_3].x_{23} + [\hat{y}_4 - y_4].x_{24}$$
$$= [0 - 0].0 + [0 - 0].1 + [0 - 0].0 + [1 - 1].1 = 0$$

$$\sum_{i=1}^{4}(\hat{y}_i - y_i) = [\hat{y}_1 - y_1] + [\hat{y}_2 - y_2] + [\hat{y}_3 - y_3] + [\hat{y}_4 - y_4]$$
$$= [0 - 0] + [0 - 0] + [0 - 0] + [1 - 1] = 0$$

$\frac{\partial E}{\partial w_1}$ =2$\sum_{i=1}^{4}(\hat{y}_i - y_i)\, x_{1i}$=2.0=0

$\frac{\partial E}{\partial w_2}$ =2$\sum_{i=1}^{4}(\hat{y}_i - y_i)\, x_{2i}$=2.0=0

$\frac{\partial E}{\partial w_0}$ =2$\sum_{i=1}^{3}(\hat{y}_i - y_i)$=2.0=0

$w_1^1 = w_1^0 - \lambda \frac{\partial E}{\partial w_1}$ =1-0.5*0=1

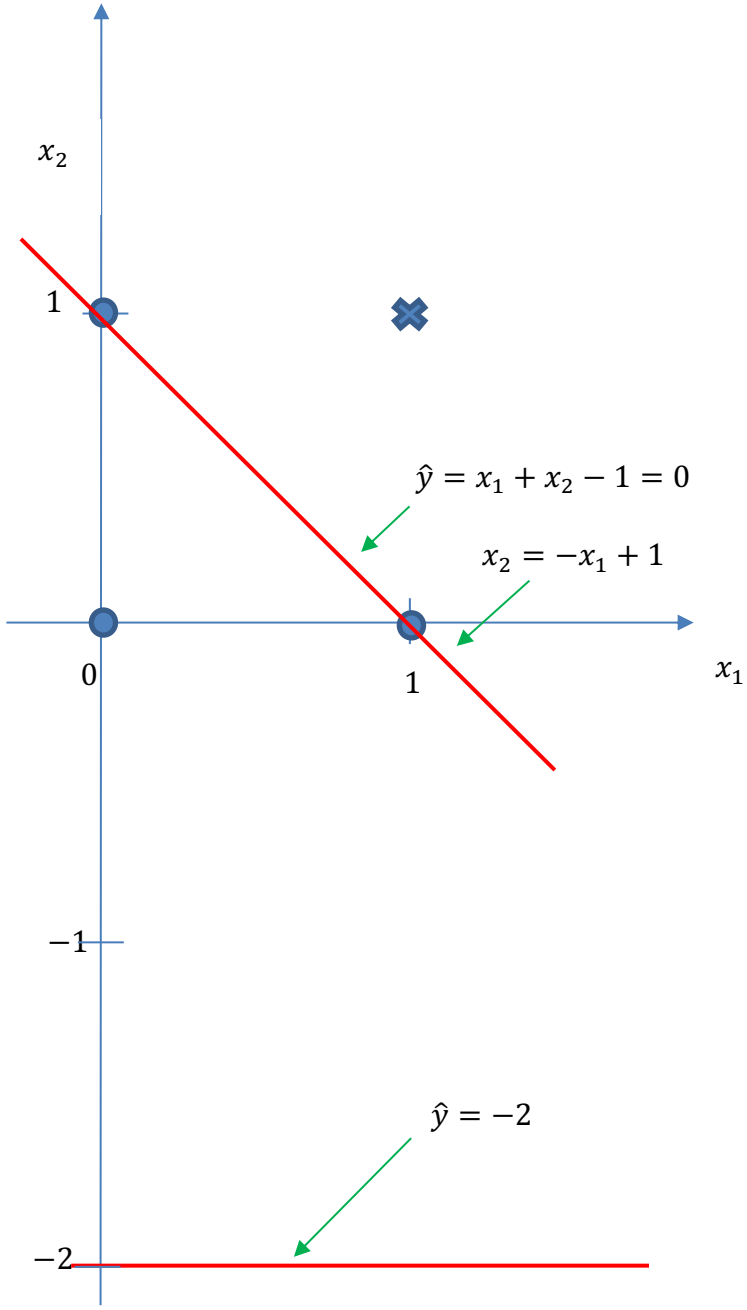
$w_2^1 = w_2^0 - \lambda \frac{\partial E}{\partial w_1}$ =1-0.5*0=1

$w_0^1 = w_0^0 - \lambda \frac{\partial E}{\partial w_1}$ =-1-0.5*0=-1

Since total error E is zero and the three weights, $w_1^1 = w_1^0$, $w_2^1 = w_2^0$ and $w_0^1 = w_0^0$. Therefore, they have been stablished.

$$\hat{y} = x_1 + x_2 - 1$$

b) Hence , plot the initial and final decision line for the classification obtained in part (a).



Initial Decision Line

$$\hat{y} = w_1 x_1 + w_2 x_2 + w_o = 0.x_1 + 0x_2 - 2$$

Final Decision Line

$$\hat{y} = x_1 + x_2 - 1$$